SAKIUL MOWLA RAMIM

Lead Embedded Firmware Engineer Chittagong, Bangladesh | +880 1836-655063 |
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PROFESSIONAL SUMMARY

Results-driven Lead Embedded Firmware Engineer with over 4 years of hands-on experience architecting, developing, and validating cutting-edge MCU-based systems. Operating a fully equipped remote hardware R&D lab, specializing in Embedded C/C++, RTOS environments, and bare-metal programming. Highly proficient in MCU register-level manipulation, custom ISR development, and hardware schematic integration for industrial and robotic systems. Proven ability to build deterministic testing frameworks utilizing logic analyzers, UART traces, and Python, ensuring real-time determinism and uncompromising reliability in complex embedded edge devices.

TECHNICAL SKILLS

- **Languages & RTOS:** Embedded C, C++, FreeRTOS, Python (Test Automation & Scripting), Bare-Metal Programming.
- **MCU Architectures & Toolchains:** ARM Cortex-M (STM32), Xtensa (ESP32), GCC ARM Toolchain, CMake, HAL/BSP Customization.
- **Hardware & Peripherals:** Register-level configuration of GPIO, UART, SPI, I2C, ADC, PWM, Timers. Deep expertise in interpreting MCU reference manuals and hardware schematics.
- **Testing & Validation:** Deterministic Testing (Logs/Traces/Timing Validation), Emulation readiness (QEMU/Renode), Logic Analyzers, Oscilloscopes, Interrupt Service Routine (ISR) debugging.
- **Connectivity & Industrial Protocols:** Modbus (RTU/TCP), RS-485, BACnet, MQTT, LoRaWAN, BLE.
- **Version Control & Hardware Design:** Git/GitHub, PCB Design (EasyEDA), Galvanic Isolation, Solar BMS.

PROFESSIONAL EXPERIENCE

NEXUS FISH NETWORK | *Remote / Hybrid* **Lead R&D Engineer (Firmware & Hardware)** | *Feb 2025 – Present Spearheading the firmware architecture for "SmartFishAuto™," an AI-driven commercial aquaculture robotics and automation ecosystem.*

- **Firmware Architecture:** Designed and implemented non-blocking Embedded C++ firmware utilizing **FreeRTOS**, managing concurrent execution of 8+ sensor polling threads and autonomous robotic motor control tasks.
- **Peripheral & ISR Management:** Configured MCU peripherals (ADC, PWM, I2C) at the register level to ensure real-time determinism. Developed custom Interrupt Service Routines (ISRs) for immediate fault detection and safety-critical laser predator defense mechanisms.
- **Hardware Bring-up & Validation:** Collaborated closely on schematic reviews to integrate firmware with custom 2-Layer PCBs. Built deterministic test harnesses using logic analyzers and UART traces to validate galvanic isolation and system timing.

- **HAL/BSP Extension:** Maintained and extended Hardware Abstraction Layers to interface complex RS-485 industrial water quality sensors smoothly with the core MCU logic.

ENERZYZ | *Remote / Contract* **Embedded Systems Engineer** | *Nov 2025 – Jan 2026*

- **Industrial Protocol Firmware:** Engineered robust C/C++ firmware modules to implement **Modbus (RTU/TCP)** and **BACnet**, bridging legacy industrial controllers with modern edge gateways (ESP32 / Orange Pi).
- **Test Automation:** Developed scalable Python scripts to emulate industrial sensor traffic, creating deterministic pass/fail criteria for firmware payload delivery under high-network-stress conditions.

PADMA SMART TECHNOLOGY LTD | *Dhaka, Bangladesh* **Embedded Firmware Engineer** | *Nov 2023 – Jan 2025*

- **Firmware Optimization:** Refactored legacy Embedded C codebases for a government smart water metering project (LoRaWAN).
- **Power Management:** Manipulated deep-sleep registers and optimized MCU clock speeds, directly improving the battery life of remote field sensors by 40% while maintaining reliable transmission windows.

GRIT TECHNOLOGIES LIMITED | *Dhaka, Bangladesh* **Embedded Systems Engineer** | *Apr 2023 – Oct 2023*

- **Industrial Reliability:** Programmed and deployed robust bare-metal firmware for vibration and temperature monitoring sensor nodes for **British American Tobacco (BAT)**.
- **Low-Level Debugging:** Resolved complex timing and concurrency issues on the RS-485 bus using oscilloscope signal validation, ensuring predictive maintenance data integrity in high-EMI industrial environments.

IOTWITHRAMIM (FREELANCE) | *International* **Freelance Embedded Developer** | *Dec 2021 – Present*

- **Memory & Audio Management:** Engineered an OEE tracking system for the Garment industry featuring highly optimized I2S audio feedback and fault-tolerant local SD Card data logging via SPI.

EDUCATION

B.Sc. in Computer Science & Engineering | *CGPA: 3.55 / 4.00 University of Creative Technology, Chittagong (UCTC)* | *2022*

Diploma in Computer Science & Engineering *Shyamoli Ideal Polytechnic Institute, Chittagong* | *2016*

ADDITIONAL EMBEDDED PROJECTS

Full technical case studies, schematics, and source code available at: sites.google.com/view/ramim-iot

- **Disaster Management Telemetry:** Engineered remote soil moisture and telemetry firmware for a Heavy Rain & Landslide Warning System deployed in vulnerable refugee zones (UN/NGO).
- **Smart Agriculture Ecosystem:** Developed Python-based computer vision integration and automated IoT irrigation logic for thesis project.

- **Automotive Telemetry Node:** Bare-metal development for a real-time multi-sensor vehicle monitor detecting impact, fire, and fuel leakage via localized ADC/GPIO polling.

(Note: Certifications in Industrial IoT (bdtask Ltd) and Web Application Development (Beacon IT) available upon request).